

Question	Answer	Marks
9(a)	emission of electrons (from a metal surface)	B1
	when electromagnetic radiation is incident (on surface / electrons)	B1
9(b)(i)	current falls to zero when applied voltage equals energy per unit charge of emitted electrons	B1
	energy of photon depends on frequency	B1
	maximum energy of electron depends on energy of photon	B1
9(b)(ii)	<i>Any three points from:</i> • threshold frequency = 1.5×10^{15} Hz • threshold wavelength = 2.0×10^{-7} m • work function = 6.2 eV (or 9.9×10^{-19} J) • Planck constant = 6.6×10^{-34} J s (not 6.63×10^{-34} J s) • number per unit time (of photons / electrons) = $1.7 \times 10^{16} \text{ s}^{-1}$ (in stage 1) • power of incident radiation = 0.028 W (in stage 1)	B3

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